



HACK2SKILL

BHARATIYA ANTARIKSH HACKATHON 2026



Team Name : Zero-Flux

Team Leader Name : Abhishek Anand

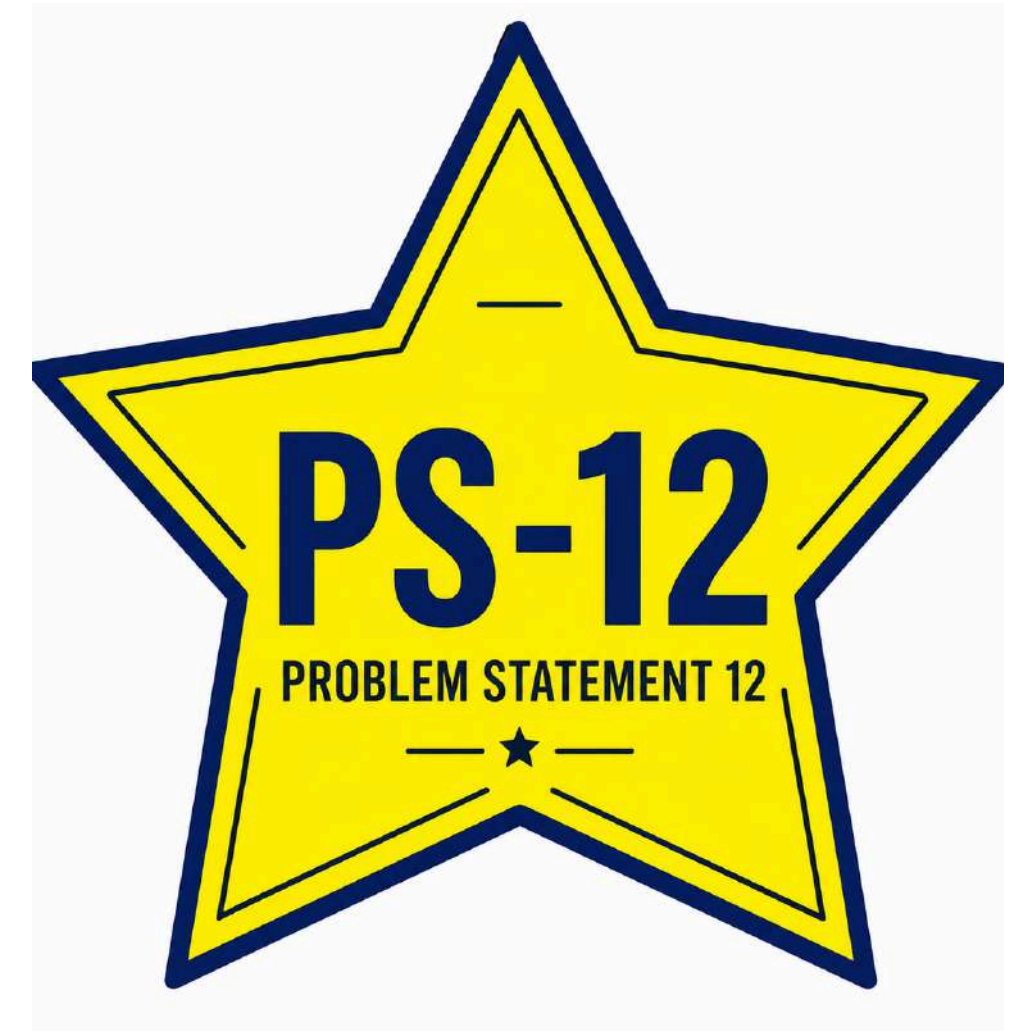
Problem Statement: Fill in the Frames Seamlessly - Enhancing Temporal Resolution of Satellite Imagery using the techniques of AI/ML based on Optical Flow

Team Zero-flux

Team Leader Name: Abhishek Anand
College : PESU, Bengaluru, Karnataka

Member-1 Aashlesh P
College: PESU, Bengaluru, Karnataka

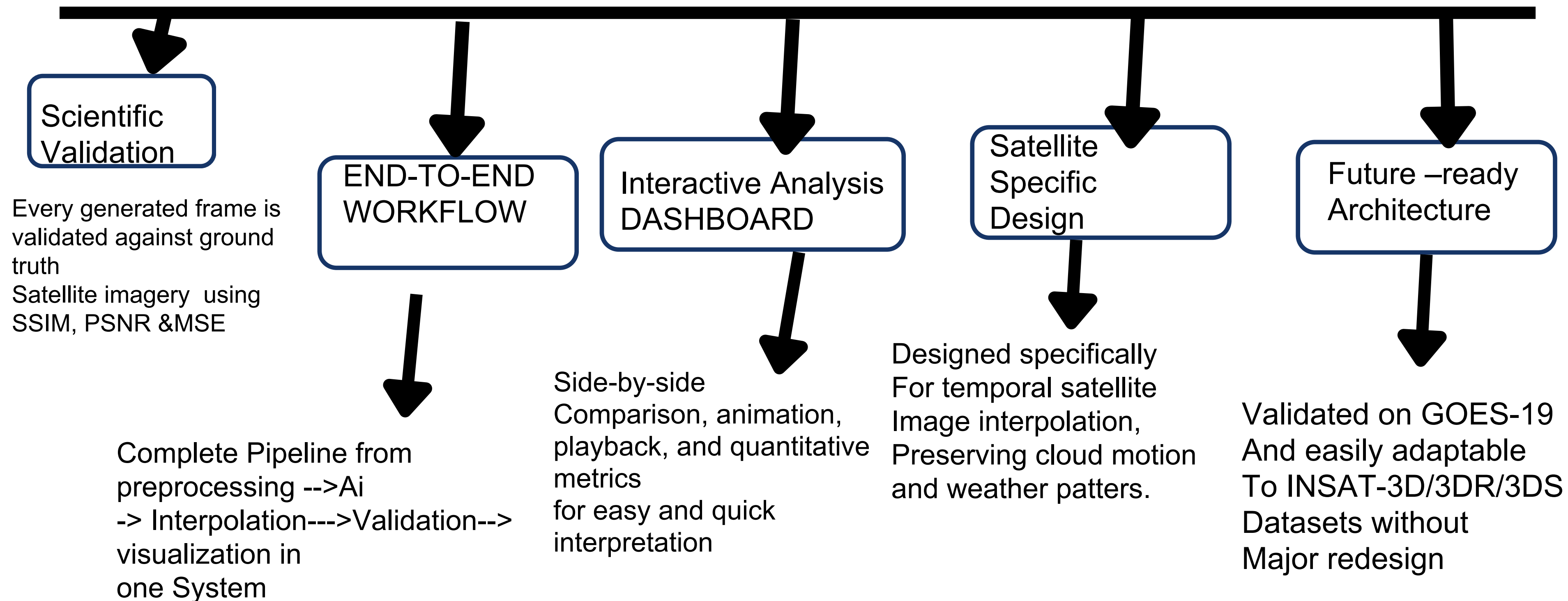
Member-1 Vidit Soni
College: PESU, Bengaluru, Karnataka



#Novelty in our solution

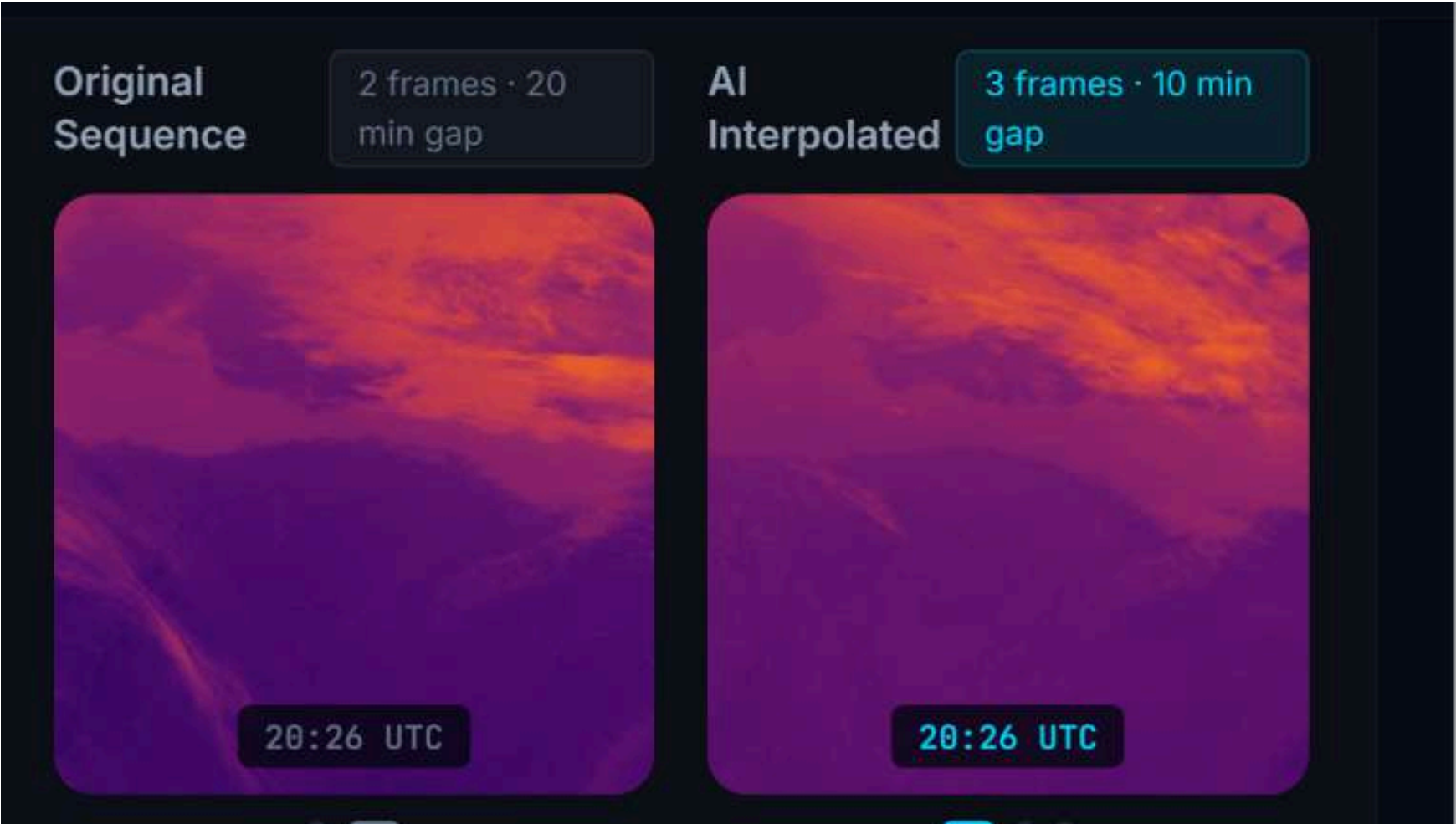
Existing Approaches	Our Approach (Zero Flux)
Generate interpolated satellite images only	Generate + scientifically validate every interpolated frame using ground-truth data
Focus mainly on visual output	Quantitative evaluation using SSIM, PSNR & MSE to measure reconstruction quality
Model-centric research with limited usability	Interactive web dashboard for visualization, comparison and analysis
Works as a standalone AI model	End-to-end pipeline: Preprocessing → AI Inference → Validation → Visualization
Limited support for operational analysis	Built for meteorological workflows with scalable integration for INSAT-3D/3DR/3DS datasets

#USP of the proposed solution

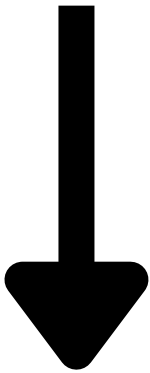


#Features Offered By us

1. Pixel-wise Difference Analysis

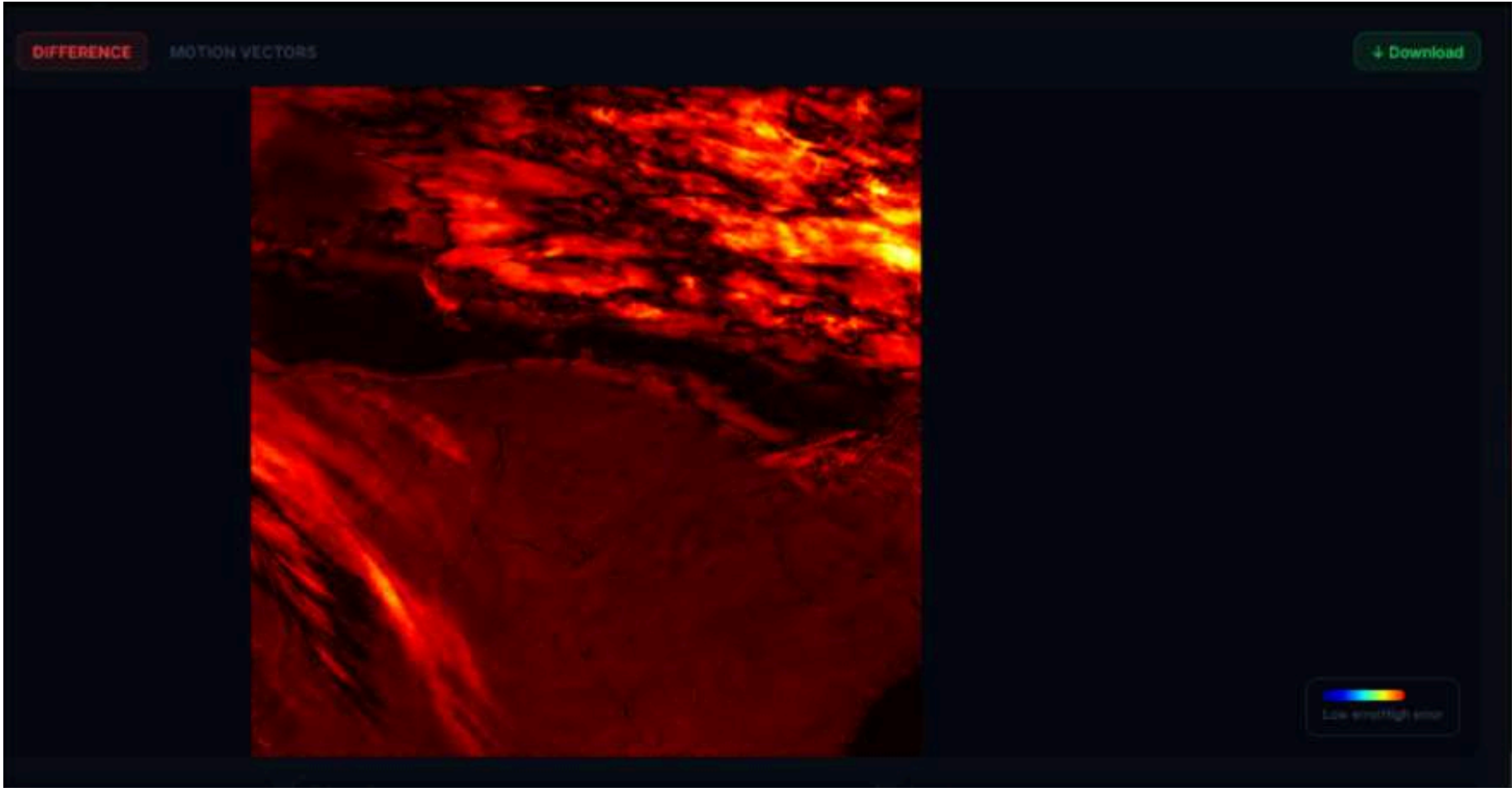


2. Quantitative Performance Evaluation

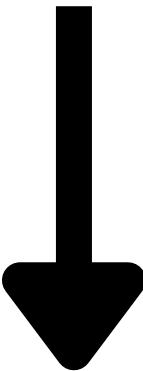


#Features Offered by Us

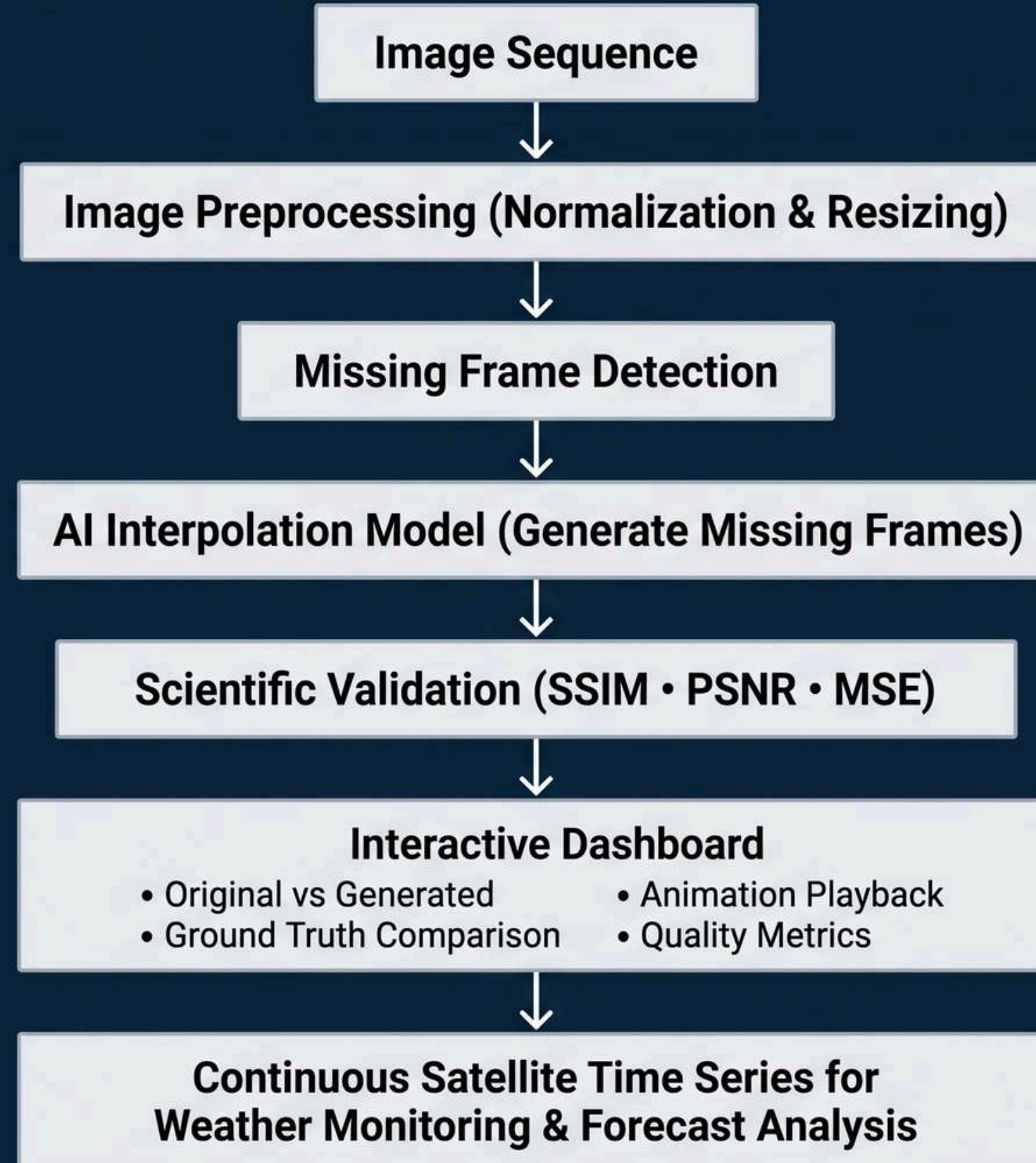
3.Ground Truth Satellite Reference



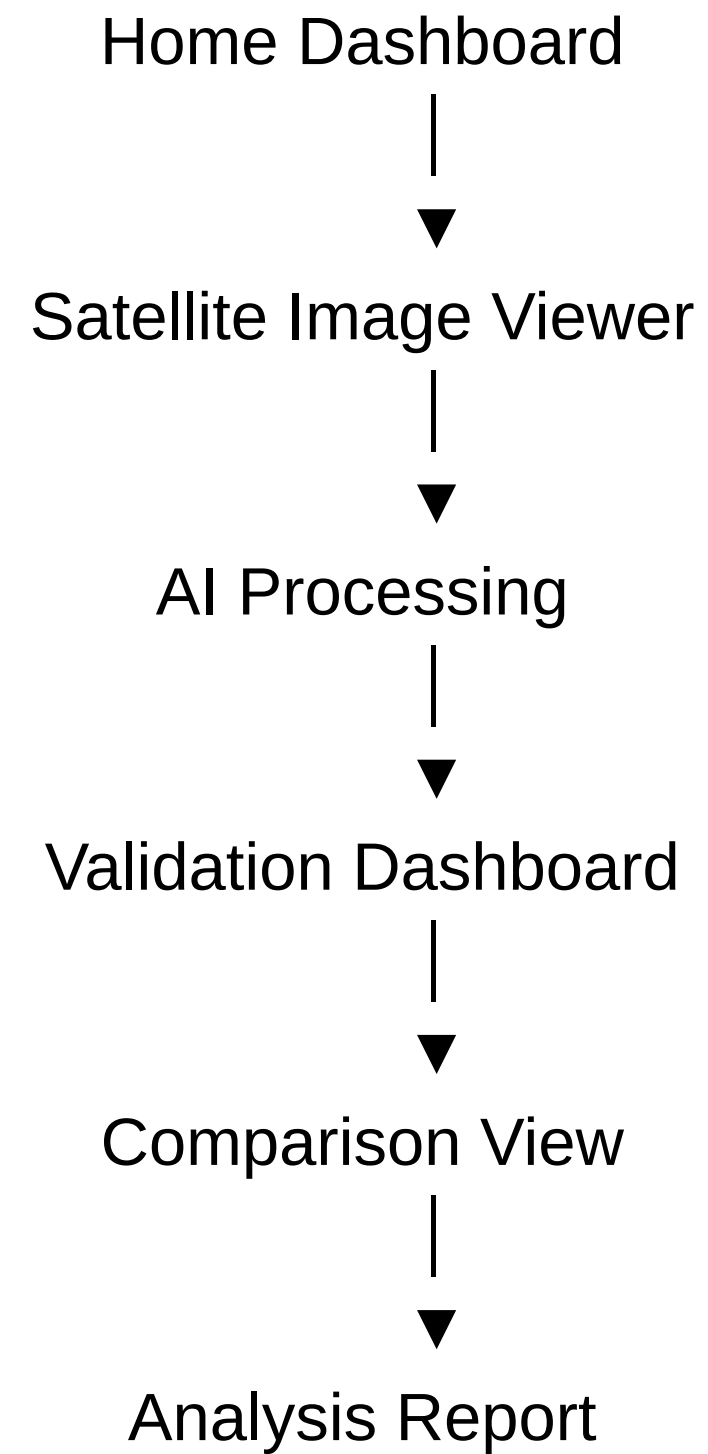
4.Performance Comparison Across Methods



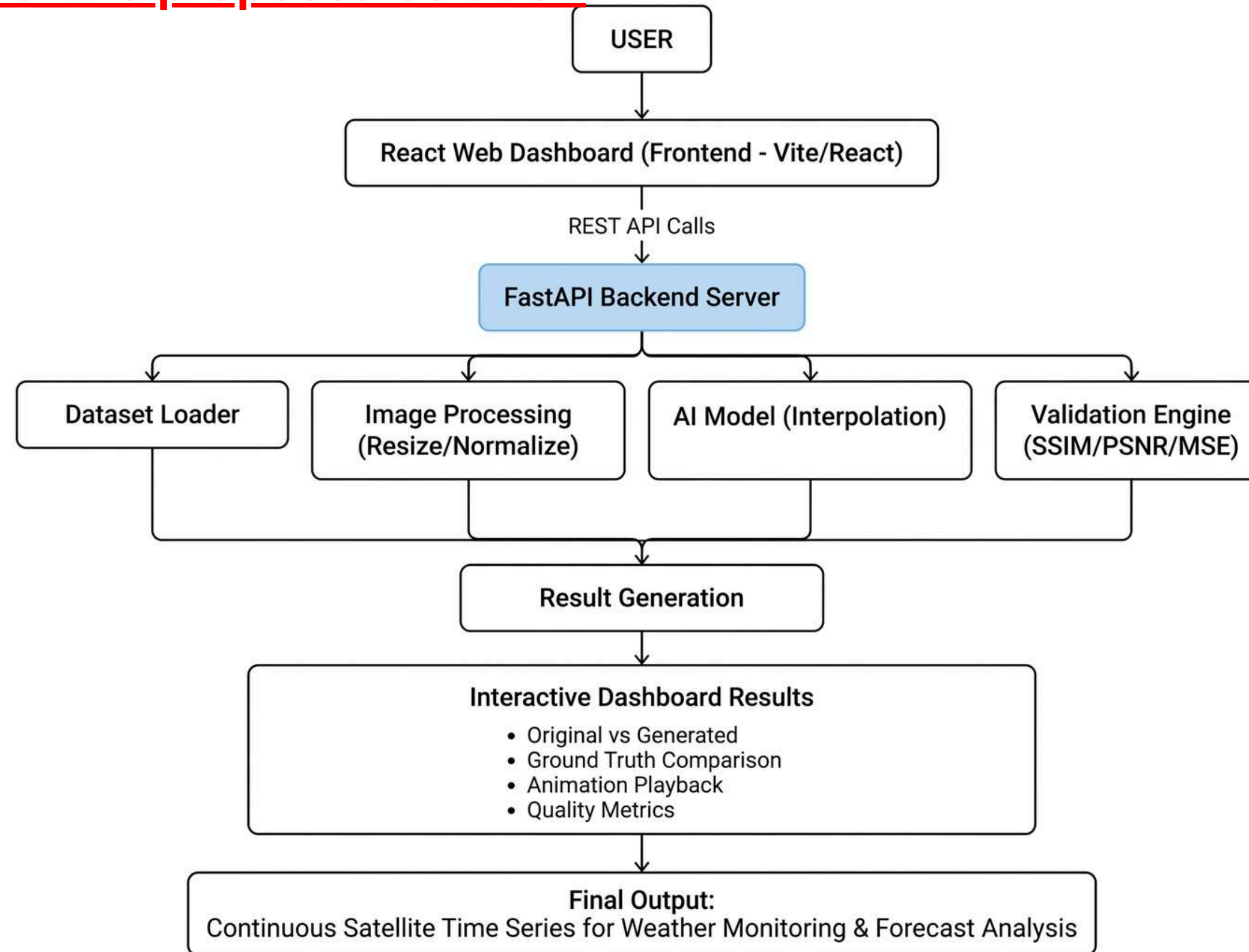
#Process flow diagram or Use-case diagram



#Mock diagrams of the proposed solution



#Architecture diagram of the proposed solution



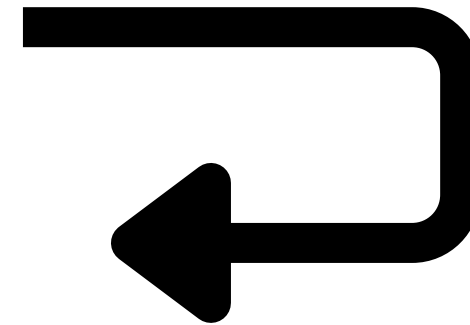
#Technologies to be used in the solution:

<u>Category</u>	<u>Technology</u>	<u>Purpose</u>
Programming Language	Python	AI model development & backend services
Deep Learning Framework	PyTorch	Satellite image interpolation model
Computer Vision	OpenCV	Image preprocessing & frame manipulation
Data Processing	NumPy, Pandas	Numerical computation & dataset handling
Backend	FastAPI	REST API for AI inference
Frontend	React (Vite)	Interactive visualization dashboard
UI Framework	Tailwind CSS	Responsive user interface
Scientific Validation	SSIM, PSNR, MSE	Quality evaluation of generated frames
Satellite Dataset (Development)	GOES-19	Model training, testing & validation
Target Dataset (Deployment)	INSAT-3D / INSAT-3DR / INSAT-3DS	Future operational deployment for Indian meteorological observations
Version Control	Git & GitHub	Source code management & collaboration

Surprise Component from Our Team

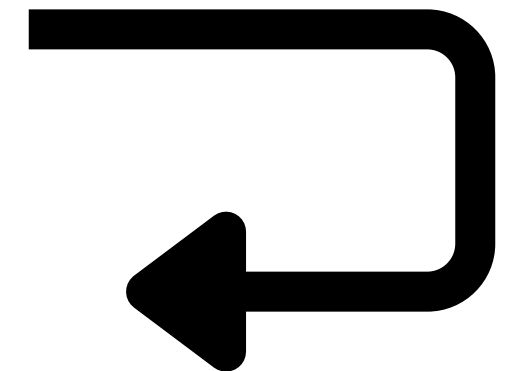
Please Check OUR Team pitch here -:

<https://www.youtube.com/watch?v=agxBvdCK8Ig>



Please check Our Github repo-:

<https://github.com/Abhishekanand19/satellite-frame-interpolation>





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THANK YOU